

Video Transcript - Part 1

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Hi everyone, welcome to part one of my AWS Certified Machine Learning Associate journey. Before we touch any AWS console or write a single line of code, I want to slow down and lay the groundwork properly. In this video, we're covering two terms you'll hear constantly throughout this course: artificial intelligence and machine learning. We'll look at what they actually mean, how they relate to each other, and where you're already seeing them in everyday life.

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Here's what we'll cover. First, a clear definition of AI. Then what machine learning actually is, and how it fits inside the bigger AI picture. After that, a quick look at how ML systems learn, the main types of machine learning, some real world applications, and finally, where things stand right now with the explosion of generative AI.

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Let's start with artificial intelligence. At its core, AI is the broad field of building machines that can perform tasks that normally require human intelligence, things like understanding language, recognizing images, making decisions, or solving problems. AI isn't one single technology. It's an umbrella term that covers everything from simple rule based systems to the advanced models powering tools like Siri, self-driving cars, and chatbots. Chances are, you've already used two or three of these today without even thinking about it.

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Machine learning is one of the main ways we actually build AI today. Instead of programming explicit rules for every situation, we give a computer data and let it find patterns on its own. The system learns from examples and improves its performance over time without being explicitly told what to do for every case. So if AI is the goal, machine learning is one of the most powerful paths to get there.

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It helps to picture this as nested circles. Artificial intelligence is the outer, biggest circle, the overall goal. Inside that sits machine learning, a specific approach to achieving AI using data driven learning. And inside machine learning sits deep learning, which uses layered neural networks to handle especially complex problems, like image recognition or natural language understanding. Every deep learning system is machine learning, and every machine learning system is a form of AI, but not the other way around.

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So how does a machine learning system actually learn? It generally comes down to four steps. First, you collect data, the raw material the model will learn from. Second, that data is used to

train a model, where the algorithm looks for patterns and relationships. Third, you evaluate the model to see how accurate its predictions are. And fourth, once it performs well, you deploy it so it can make predictions on brand new, unseen data. This cycle usually repeats, since models get retrained as new data comes in.

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Machine learning itself comes in a few different flavors. Supervised learning uses labeled data, meaning the correct answer is already known, and it's great for tasks like predicting prices or classifying emails as spam. Unsupervised learning works with unlabeled data and looks for hidden patterns or groupings on its own, useful for customer segmentation. And reinforcement learning learns through trial and error, getting rewarded for good decisions, which is how systems learn to play games or control robots.

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You're probably interacting with machine learning more than you realize. In healthcare, it helps detect diseases earlier from medical scans. In finance, it flags fraudulent transactions in real time. Retailers use it to recommend products you might like. Transportation companies use it for route optimization and self-driving technology. Streaming platforms use it to recommend your next show. And manufacturers use it to predict when equipment needs maintenance before it breaks down. Almost every industry you can think of has found a way to put it to work.

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Right now, we're in a particularly exciting moment for this field. The rise of generative AI and large language models, like the ones behind tools such as ChatGPT and Claude, has pushed machine learning into the mainstream. These models can write, summarize, generate images and audio, explain code, and hold natural conversations. The global AI market is growing rapidly, and companies across every industry are racing to build ML into their products, which means the skills we're building in this course are more relevant right now than they've ever been.

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This is exactly why cloud platforms like AWS matter so much in this space. AWS gives developers and businesses the infrastructure and tools to build, train, and deploy machine learning models without having to manage physical hardware or build everything from scratch. Services like Amazon SageMaker handle the heavy lifting of the ML lifecycle we just talked about, from data preparation to training to deployment. That's exactly what this course is going to dive into over the coming weeks.

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So to recap, AI is the big picture goal of building intelligent systems, machine learning is the data driven approach we use to get there, and it's already showing up across healthcare, finance,

retail, and more. In the next videos, we'll go deeper into specific ML concepts and start getting on with AWS tools. Thanks for watching, and I'll see you in the next one.